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A unified representation and fusion framework of multi-source heterogeneous data for fault diagnosis in industrial processes

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ABSTRACT

In the increasingly integrated and complex industrial processes, it is difficult for a single type of data to adequately characterize faults due to the correlation and multi-source heterogeneity of process data. Therefore, to achieve accurate fault diagnosis, multi-source heterogeneous data is necessary to be integrated to obtain comprehensive and multi-scale fault information. However, significant differences in dimensions and structures are exhibited by multi-source heterogeneous data, which may make the high-quality data fusion difficult to achieve. Responding to the above issues, in order to achieve a unified representation and fusion of multi-source heterogeneous data, and enhance the fault diagnosis performance, a unified representation and fusion framework is proposed for multi-source heterogeneous data based fault diagnosis in industrial processes. Specifically, the optimal region containing useful information in the fault image data is first selected as the representative region. Subsequently, tensor fusion is performed on the time series data and the selected region of image data respectively, and the low-rank decomposition is used to obtain the low-dimensional unified vector representation for data fusion. Furthermore, a fault diagnosis module based on the multi-scale lightweight convolutional neural network is constructed in which the multi-scale features are extracted and fused from the fusion vector and then used for fault diagnosis. Finally, to validate the effectiveness and superiority of the proposed framework, comparative experiments based on actual data from the hot rolling process are conducted.

1. Introduction

In the modern industrialization processes, with the continuous developments of production technologies and the increasing demands for production, industrial processes are gradually advancing towards automation, intelligence, and integration, which greatly enhances the corresponding productivity and product quality. Associated with those trends, the risk of faults occurring and the difficulty of fault diagnosis are also increased. Modern industrial processes typically rely on abundant sensors and complex control systems, which often have strong coupling characteristics. Once a fault occurs, it may propagate through the coupling relationships between systems, leading to multiple faults, production stagnation, and thus severely affecting the safety, stability, and reliability. By means of advanced fault diagnosis technologies, faults in the industrial processes can be promptly and accurately identified and addressed, effectively ensuring the continuity and security of industrial operations. Therefore, as an important technology to promote the development of industrial processes, fault diagnosis has a wide application prospect [1,2].

In recent years, due to the advancements of real-time monitoring and data acquisition technologies, large amounts of data generated in industrial processes are collected and stored, providing rich sources of information for fault diagnosis. Therefore, data based fault diagnosis methods have gradually become mainstream. Those methods automatically identify and diagnose the faults in industrial processes by analysing massive data, extracting features from them, and identifying potential fault patterns. Currently, data based fault diagnosis methods typically include machine learning based methods and deep learning based ones. Machine learning based methods, such as multivariate statistics [3], artificial neural networks (ANN) [4], random forest (RF) [5,6], support vector machine (SVM) [7,8], and k-nearest neighbour (KNN) [9], are able to construct and train models based on historical data to identify and predict the fault patterns. However, as the industrial processes become increasingly complex, those fault diagnosis models may struggle to adapt when processing fault data with large-scale, high-dimensional, and nonlinear, potentially leading to insufficient adaptability and affecting the fault diagnosis accuracy.

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By comparison, deep learning techniques are gradually applied in the field of fault diagnosis, effectively solving the problem of fault diagnosis in complex scenarios [10,11]. Deep learning based ones, such as auto encoder (AE) [12–14], convolutional neural network (CNN) [15–17], deep belief network (DBN) [18,19], and generative adversarial network (GAN) [20–22], can effectively extract deep features of data through constructing multi-layer networks to capture information that can better represent fault patterns. Compared with the machine learning based ones, those methods significantly improve the ability to analyse the data with high-dimension and complex nonlinear relationships, and possess stronger feature extraction and learning capabilities. Although the above methods perform well in fault diagnosis, they generally concentrate on analysing and modelling a single type of data. However, in the increasingly complex industrial processes, a fault often leads to abnormal changes in multiple types of data. Accordingly, the loss of fault information may be caused by analysing only a single type of data, affecting the fault diagnosis accuracy. To fully utilize the complementary and multi-scale information from different types of data and prevent information loss, multi-source heterogeneous data fusion based methods are gradually implemented in the fault diagnosis fields for industrial processes.

For the multi-source heterogeneous data fusion based fault diagnosis methods, comprehensive and multi-scale information can be obtained through integrating fault data from different sources, overcoming the limitations of a single type of data, and characterizing faults more comprehensively. According to different ways of fusion, fault diagnosis methods based on multi-source heterogeneous data fusion can be categorized into decision level fusion, feature level fusion, and data level fusion. Decision level fusion [23–25] refers to independently processing and analysing the data from different sources to obtain individual diagnosis results, which are then fused to obtain the final results. Those methods can flexibly select different decision fusion strategies. Meanwhile, by combining multiple decision results, the impact of single model decision error can be effectively reduced, improving the robustness of fault diagnosis model. However, the overall performance often depends on the fault diagnosis accuracy of each individual model. Additionally, during the decision level fusion, the details of original data are often overlooked, which may lead to key information loss. Furthermore, as the number of data sources grows, the model complexity is also increased, leading to significant computational resources and time consumption, and affecting the efficiency of fault diagnosis.

Feature level fusion based methods [26–28] firstly extract features from data originating from different sources, which are then fused through concatenation, weighted summation, or other ways. Subsequently, the fused feature vectors are analysed for fault diagnosis. Those methods are able to perform feature extraction independently for each data source, and achieve dimensionality reduction of original data while acquiring features, thus reducing the computational complexity of feature fusion process to a certain extent. Meanwhile, through independent feature extraction, a unified feature representation of heterogeneous data can be obtained. However, the effectiveness of feature level fusion highly depends on the feature extraction quality. The insufficient feature extraction may lead to information loss, significantly reducing the effectiveness of feature fusion. Furthermore, to achieve adequate feature extraction, the structure of networks is usually more complex, which tends to increase the overall model complexity and affects the fault diagnosis efficiency.

Data level fusion based methods [29,30] usually fuse raw data from different sources through concatenation, weighted averaging or other ways, and then feature extraction is performed on the fused data to obtain features for fault diagnosis. Compared with the decision and feature level fusion based methods, the data level fusion based ones, to some extent, can retain the fault information contained in each data source and avoid fault information loss in the feature extraction processes. Additionally, to perform feature extraction and fault diagnosis, the decision and feature level fusion based methods usually require

constructing models for each data source, whereas the data level fusion based methods only require modelling the fused data, thus simplifying the model and training processes. However, in practical industrial processes, raw fault data is usually high dimensionality and large amount, leading to high model complexity when the data fusion is performed directly. Furthermore, due to the diversity of data types, there are often significant differences in structures and dimensions, making it difficult to fuse and collaboratively analyse the heterogeneous data.

Based on the above analysis, it can be seen that the decision and feature level fusion based methods can realize heterogeneous data fusion, while they are prone to information loss. Alternatively, the data level fusion based methods can effectively retain the information from the original data, but they may face challenges in achieving effective fusion of heterogeneous data. Therefore, in order to achieve high-quality multi-source heterogeneous data fusion and avoid fault information loss, in this paper, a unified representation and fusion framework of multi-source heterogeneous data is proposed for fault diagnosis in industrial processes. It constructs a unified and standardized representation method to eliminate the structural differences between multi-source heterogeneous data, enabling them to be collaboratively analysed within a unified framework, thereby effectively preventing information loss and improving the ability of data integration. Furthermore, to address the common problem of high model complexity in multi-source heterogeneous data fusion fault diagnosis methods, lightweight methods are adopted to reduce model parameters. The specific innovations are highlighted as follows:

- a fault image data optimal region selection module based on the self-attention mechanism is constructed, which selects the region containing valid information in fault image data as the representative one, aiming at lowering the model complexity while reducing redundant information in the fault image data;
- a low-rank decomposition based unified representation and fusion method is proposed, in which the tensor fusion is performed on multi-source heterogeneous data to prevent information loss, and the low-rank decomposition is utilized to obtain a low-dimensional unified vector representation of the fused tensor, so as to overcome the challenge of fusing heterogeneous data;
- a multi-scale convolution based lightweight CNN module is constructed for fault diagnosis, which utilizes multiple convolution kernels with different sizes to extract features and replaces traditional convolution with depthwise separable convolution to improve the fault feature extraction capability and reduce the model complexity;
- two cases based on a representative hot rolling process (HRP) are set up to verify the effectiveness and superiority of the proposed framework.

The remaining sections of the paper are structured as follows. Principles of low-rank weighted decomposition and depthwise separable convolution are introduced in Section 2. The unified representation and fusion framework of multi-source heterogeneous data for fault diagnosis is detailed in Section 3. After that, two cases are set up in Section 4. Finally, conclusions and some discussions of future work are provided in Section 5.

2. Background and problem formulation

In this section, principles of low-rank weighted decomposition and depthwise separable convolution are briefly introduced, followed by the formulation of the problem.

2.1. Low-rank weighted decomposition

Low-rank weighted decomposition is able to reconstruct a matrix or tensor into the matrix of multiple low-rank factors [31]. For a order- $(K + 1)$ tensor $\mathcal{Z}$ with dimension $d_1 \times d_2 \times \dots \times d_K \times d_h$, it can be viewed as d_h order- K tensors, where each order- K tensor can be represented as $\tilde{\mathcal{Z}}_j, j = 1, 2, \dots, d_h$ with dimension $d_1 \times d_2 \times \dots \times d_K$. For

each order- K tensor $\tilde{Z}_j$, there exists a way to decompose it into vectors:

$$\tilde{Z}_j = \sum_{i=1}^R \bigotimes_{k=1}^K \alpha_{k,j}^{(i)} \quad (1)$$

where $\bigotimes$ is the outer product operation, and R is the minimal rank of the tensor that makes the decomposition valid. The set of vectors $\{\{\alpha_{k,j}^{(i)}\}_{k=1}^K\}_{i=1}^R$ is decomposition factors with the rank R of original tensor. Those vectors can be recombined into K low-rank factors. Let $z_k^{(i)} = [\alpha_{k,1}^{(i)}, \alpha_{k,2}^{(i)}, \dots, \alpha_{k,d_h}^{(i)}]$, the low-rank tensor $\mathcal{Z}$ can be reconstructed through the K specific low-rank factors:

$$\mathcal{Z} = \sum_{i=1}^R \bigotimes_{k=1}^K z_k^{(i)} \quad (2)$$

Through low-rank weighted decomposition, high-dimensional tensors can be decomposed into multiple low-rank factors, where each low-rank factor corresponds to a dimension of tensor. In modern industrial processes, the multi-source heterogeneity of data makes it difficult to fully represent fault information for a single data source. Moreover, the dimensions and structures of data from different sources are very different, which leads to difficulties in achieving data fusion through simple concatenation. Tensor fusion methods can achieve heterogeneous data fusion by means of some operations such as the outer product, while minimizing information loss as much as possible. However, due to the high dimensionality, it is usually difficult to carry out the subsequent fault diagnosis process. The low-rank weighted decomposition method can decompose the high-dimensional tensor into a low-dimensional space, which can effectively solve the problem of high tensor dimensionality. Therefore, a unified representation and fusion framework for multi-source heterogeneous data is constructed by combining tensor fusion with low-rank weighted decomposition methods, aiming at multi-source heterogeneous data fusion.

2.2. Depthwise separable convolution

As a lightweight convolution method for CNN, depthwise separable convolution [32] aims to reduce the model complexity by decommissioning traditional convolution operations and reducing the number of calculations and parameters.

Deep separable convolution is composed of two stages, depthwise convolution and pointwise convolution. In the depthwise convolution stage, each channel of the input data corresponds to a convolution kernel. For each convolution kernel, the convolution operation is performed only on the data of the corresponding channel. Therefore, the same number of output channels is available as input channels. The computational cost and the parameter number of depthwise convolution can be expressed by:

$$N_c^D = K \times K \times C_{in} \times H \times W \quad (3)$$

and

$$N_p^D = K \times K \times C_{in} \quad (4)$$

where N_c^D and N_p^D are the computational cost and the parameter number of depthwise convolution, respectively. $K \times K$ is the size of convolution kernel, and C_{in} , H , and W are the number of channels, height and width of input data, respectively.

By performing depthwise convolution on the input data, C_{in} output features can be obtained. However, when there are fewer input channels, the output feature channels are also restricted, which affects the ability to express features of the model. Pointwise convolution can perform the linear combination on features output from depthwise convolution, thereby enhancing the feature representation capability of model by increasing the number of output channels. Pointwise convolution performs convolution operation on the output features

from depthwise convolution with 1×1 convolution kernels, and its computational cost and parameter number can be expressed by:

$$N_c^P = C_{in} \times C_{out} \times H \times W \quad (5)$$

and

$$N_p^P = C_{in} \times C_{out} \quad (6)$$

where N_c^P and N_p^P are the computational cost and parameter number of pointwise convolution, and C_{out} is the number of output channels.

The decomposition of traditional convolution can be achieved by the deep convolution and point convolution operations of depthwise separable convolution:

$$Ratio_c = \frac{(K \times K + C_{out}) \times C_{in} \times H \times W}{K \times K \times C_{in} \times C_{out} \times H \times W} = \frac{1}{C_{out}} + \frac{1}{K^2} \quad (7)$$

$$Ratio_p = \frac{(K \times K + C_{out}) \times C_{in}}{K \times K \times C_{in} \times C_{out}} = \frac{1}{C_{out}} + \frac{1}{K^2} \quad (8)$$

Compared with the traditional convolution, its computational costs and number of parameters can be effectively reduced, thereby reducing the model complexity. In practical industrial processes, fault data is often multi-source heterogeneous. The depthwise convolution operation is able to extract effective features from the data of different sources, while the pointwise convolution operation helps capture the coupling relationships between the data from different sources. Therefore, in this paper, to achieve effective feature extraction for multi-source heterogeneous data and reduce the model complexity, depthwise separable convolution is utilized to build a feature extraction and fault diagnosis module.

3. The proposed fault diagnosis framework

In this section, the proposed unified representation and fusion framework of multi-source heterogeneous data for fault diagnosis in industrial processes is introduced in detail.

3.1. Self-attention mechanism based optimal region selection for fault image data

In the fault image data collected from practical industrial processes, key fault features are often located in the specific regions of image, while the background area accounts for a large portion and does not contain fault information. If the whole fault image data is analysed, it may be interfered by background information, affecting the fault diagnosis accuracy and increasing the model complexity. Therefore, as shown in Fig. 1, the proposed method, firstly, selects an appropriate sliding window size through the genetic algorithm. Subsequently, it slides over the image data with a specific step size to divide the image into patches. Suppose that p is the total number of image patches, the image patches can be represented as $P_i (i = 1, 2, \dots, p)$. To obtain the image data regions containing important information, each image patch is transformed in dimension from a two-dimensional to one-dimensional vector and sequentially input into the self-attention mechanism based module. The query and key vectors for each patch are then calculated, which can be represented by:

$$\begin{cases} q_i = W_Q \hat{P}_i \\ k_i = W_K \hat{P}_i \end{cases} \quad (9)$$

where $\hat{P}_i$ is the one-dimensional vector obtained by flattening the i th image patch, q_i and k_i are the query vector and key vector obtained by inputting $\hat{P}_i$ into the self-attention mechanism based module, respectively, W_Q and W_K are the weight matrices of the self-attention mechanism based module.

The query vectors q_i and key vectors k_i corresponding to each image patch are concatenated to form the query matrix $Q = (q_1, q_2, \dots, q_p)^T$

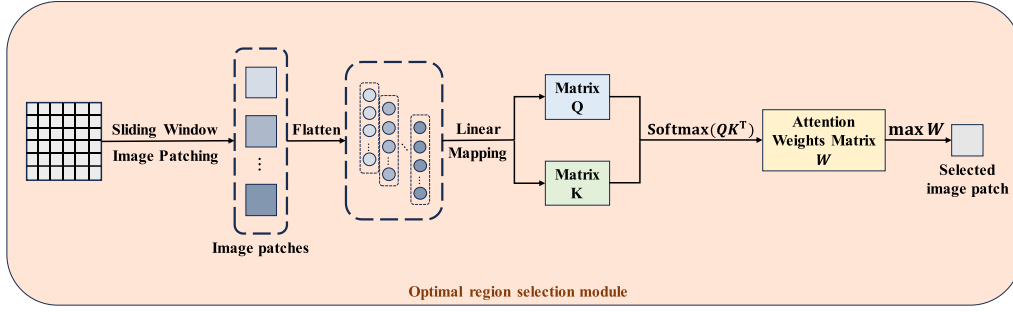


Fig. 1. Flowchart of optimal region selection module.

and the key matrix $K = (k_1, k_2, \dots, k_p)^T$ for all image patches. Subsequently, the importance weight matrix corresponding to all image blocks is calculated. The specific process can be given by:

$$W = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \quad (10)$$

where $W = (w_1, w_2, \dots, w_p)$ is the weight matrix of all image patches and $w_i (i = 1, 2, \dots, p)$ is the weight vector of each image patch, d is the size of the one-dimensional vector which each image patch is flattened into, and $\text{Softmax}(\cdot)$ represents the Softmax activation function.

After the importance weight matrix is obtained, the image patch with the highest weight is regarded as the image region containing important information. The process is shown as:

$$z^i = \hat{P}_{\arg \max(W)} \quad (11)$$

where z^i is the optimal image patch of image data from the current data source, and $\arg \max(\cdot)$ is the index of the location of the maximum value. The region z^i is selected as the representative area of image data, thereby reducing the interference from irrelevant information, and decreasing redundant computations in the data fusion process.

3.2. Low-rank decomposition based unified representation and fusion of multi-source heterogeneous data

When fusing the data from multiple time series and image data sources, it is challenging to achieve data fusion of heterogeneous data due to their different dimensions. Tensor fusion is able to obtain a high-dimensional tensor representation of data from different data sources by outer product, thus realizing the fusion of heterogeneous data. However, in actual industrial processes, time series and image data often have high dimensions, which may lead to dimensionality explosion after tensor fusion, affecting the subsequent fault diagnosis process. Therefore, as shown in Fig. 2, to address the above issues, the low-rank decomposition is used to obtain the low-rank factor matrix corresponding to each dimension (i.e., each data source) of the high-dimensional fusion tensor, which is then utilized to perform a linear transformation on it, thereby obtaining a unified low-dimensional vector representation of high-dimensional fusion tensor. Specifically, the optimal region selection module is first used to perform region selection on the image data from each image data source, obtaining the representative region for each image data source. Subsequently, tensor fusion is then performed on the data from multiple time series and image data sources, respectively.

Assuming that the amount of time series data sources is N and that of image data sources is M , the fusion process of time series and image data can be represented by:

$$\begin{cases} Z^t = \bigotimes_{n=1}^N z_n^t \\ Z^i = \bigotimes_{m=1}^M z_m^i \end{cases} \quad (12)$$

where Z^t is the tensor representations after fusion of time series data, while Z^i is the tensor representations after fusion of image data. z_n^t is

the time series data from the n th time series data source, and z_m^i is the image data from the m th image data source.

At this point, the tensor representations of time series and image data are input to the linear layers $f(\cdot)$ and $g(\cdot)$ to obtain the unified representations of them. This process can be got by:

$$\begin{cases} h^t = f(Z^t; \mathcal{W}^t; b^t) = \mathcal{W}^t \cdot Z^t + b^t \\ h^i = g(Z^i; \mathcal{W}^i; b^i) = \mathcal{W}^i \cdot Z^i + b^i \end{cases} \quad (13)$$

where h^t , h^i are the vector representations obtained by passing the tensor representations of time series and image data through the linear layers, $\mathcal{W}^t$, $\mathcal{W}^i$ are the high-order weight tensors, b^t and b^i are the biases.

According to the principle of low-rank weighted decomposition, the low-rank decomposition of rank r for $\mathcal{W}^t$ and $\mathcal{W}^i$ is respectively performed to obtain the low-rank factor matrix corresponding to each dimension of tensor Z^t and Z^i . In this paper, the bias term is set to 0. Then, the process in Eq. (13) can be expressed by:

$$\begin{cases} h^t = \left(\sum_{s=1}^r \bigotimes_{n=1}^N w_n^{t,(s)} \right) \cdot Z^t \\ = \left(\sum_{s=1}^r \bigotimes_{n=1}^N w_n^{t,(s)} \cdot \bigotimes_{n=1}^N z_n^t \right) \\ = \bigwedge_{n=1}^N \left(\sum_{s=1}^r w_n^{t,(s)} \cdot z_n^t \right) \\ h^i = \left(\sum_{s=1}^r \bigotimes_{m=1}^M w_m^{i,(s)} \right) \cdot Z^i \\ = \left(\sum_{s=1}^r \bigotimes_{m=1}^M w_m^{i,(s)} \cdot \bigotimes_{m=1}^M z_m^i \right) \\ = \bigwedge_{m=1}^M \left(\sum_{s=1}^r w_m^{i,(s)} \cdot z_m^i \right) \end{cases} \quad (14)$$

where $w_n^{t,(s)}$, $w_m^{i,(s)}$ are the low-rank factor matrices corresponding to $\mathcal{W}^t$, $\mathcal{W}^i$, respectively, and $\bigwedge_{n=1}^N (\cdot)$ is the element-wise product operation.

After the above operations, the unified vector representations h^t and h^i of multi-source time series and image data are obtained. To further simplify the data fusion process and improve the stability and robustness of the model, element-wise summation is used to replace the original element-wise product in Eq. (14), thereby obtaining the unified vector representation of multi-source heterogeneous data. Meanwhile, to preserve the integrity of fault information, the unified vector representations h^t and h^i are summed to achieve the multi-source heterogeneous data fusion, which can be represented by:

$$\begin{cases} h^t = \sum_{n=1}^N \left(\sum_{s=1}^r w_n^{t,(s)} \cdot z_n^t \right) \\ h^i = \sum_{m=1}^M \left(\sum_{s=1}^r w_m^{i,(s)} \cdot z_m^i \right) \end{cases} \quad (15)$$

and

$$h = h^t + h^i \quad (16)$$

where h is the vector representation of fused multi-source heterogeneous data, integrating the information with different scales from image data and time series data.

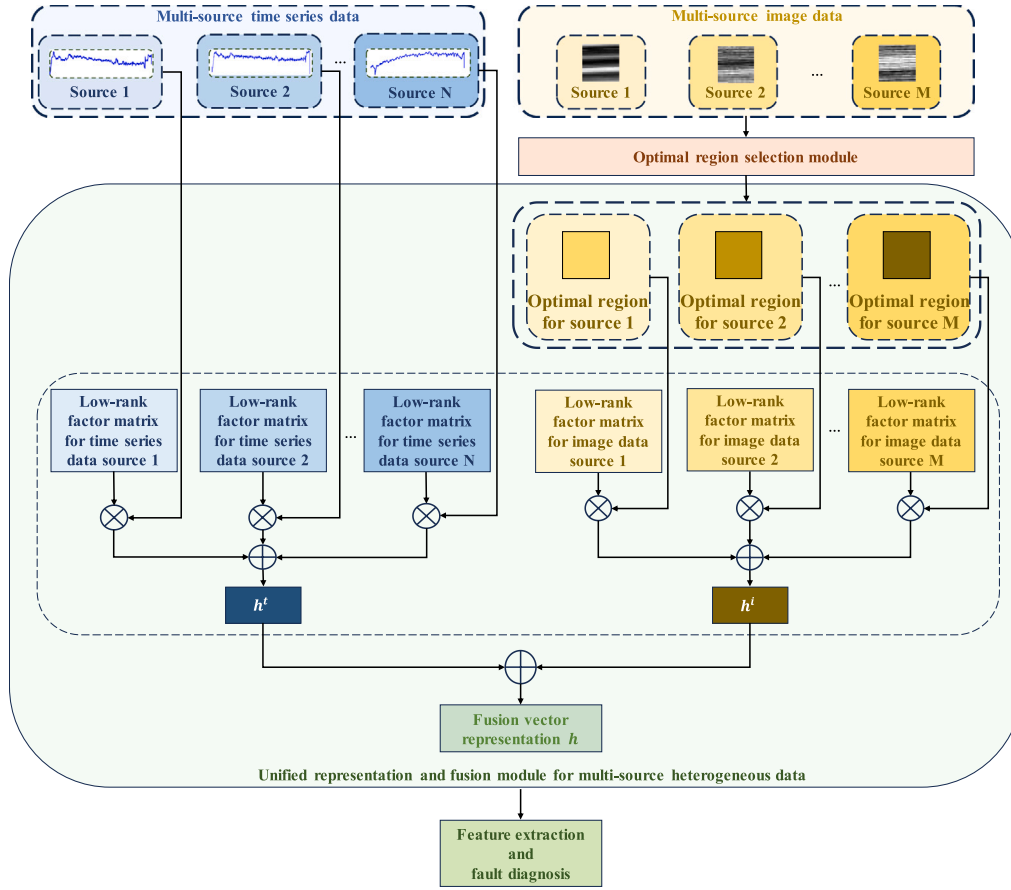


Fig. 2. Flowchart of unified representation and fusion module of multi-source heterogeneous data.

3.3. Multi-scale convolution based lightweight CNN for fault diagnosis

The fused vector h contains multi-scale information from the original image data and time series data. However, traditional single-scale convolutions struggle to capture the multi-scale fault feature information in the fused vector. To address this issue, a multi-scale convolution based lightweight CNN is constructed, enabling multi-scale feature extraction from the fused vector for fault diagnosis. As shown in Fig. 3, the model performs convolution operations on the fused vector using multiple convolution kernels of different sizes $S_u \times 1, u = 1, 2, \dots, U$, where each convolution module is implemented through depthwise separable convolutions. First, the depthwise convolution is performed on each channel of h in parallel using the convolution kernels with different scales, which can be represented by:

$$f_c^{(u)} = h_c * W_c^{(u)}, c = 1, 2, \dots, C_{in} \quad (17)$$

where h_c is the c th channel of fused vector h , $f_c^{(u)}$ is the feature vector obtained by applying the u th scale convolution kernel to the c th channel of fused vector h through depthwise convolution, $W_c^{(u)}$ is the weights of the u th scale convolution kernel corresponding to the c th channel, and $*$ is the one-dimensional convolution operation.

Through the depthwise convolution, the feature vectors with different scale of each channel in h are obtained. The pointwise convolution is then applied to reflect the coupling relationships of data between different channels, enhancing the fault feature representation capability:

$$\hat{f}_c^{(u)} = f_c^{(u)} * W_c^{(u)}, C = 1, 2, \dots, C_{out} \quad (18)$$

where $\hat{f}_c^{(u)}$ is the feature vector obtained by applying pointwise convolution to $f_c^{(u)}$, $W_c^{(u)}$ is the weight matrix of the convolution kernel with size of 1×1 .

Furthermore, the multi-scale feature vectors are concatenated to obtain the multi-scale fused feature $F = (\hat{f}_c^{(1)}, \hat{f}_c^{(2)}, \dots, \hat{f}_c^{(U)})$. Subsequently, a multi-layer fully connected network is constructed, whose output is mapped using the Softmax activation function, serving as the fault classifier. Fault diagnosis is finally performed using the multi-scale fused feature F that can be represented by:

$$\hat{y}_j = \text{Softmax}(\omega_2 \cdot (\text{ReLU}(\omega_1 \cdot F + b_1)) + b_2), j = 1, 2, \dots, \mathcal{N} \quad (19)$$

where $\hat{y}_j$ is the probability output by the fault diagnosis model that the current sample belongs to each fault class, $\text{ReLU}(\cdot)$ is the ReLU activation function, ω_1 and ω_2 are the weight matrices of fully connected layer, b_1 and b_2 are the bias term of fully connected layer, and $\mathcal{N}$ is the number of fault classes.

The structure of the model in this paper is composed of the optimal region selection module, the unified representation and fusion module, and the multi-scale feature extraction and fault diagnosis module. Firstly, historical data is used to train the model, and the process is shown in Algorithm 1. Then, the trained model is used for fault diagnosis on real-time data in industrial processes. The overall procedures of fault diagnosis are as follows:

Step 1 : The optimal region containing valid fault information in the image data is selected by the optimal region selection module of fault image data, preventing interference from background or other irrelevant information;

Step 2 : The unified representation and fusion module is used to obtain a unified vector representation of multi-source heterogeneous data, and then the fused vector representation is realized by further summing them;

Step 3 : The fused vector representation is input into the multi-scale feature extraction and fault diagnosis module, where the multi-scale feature extraction and fusion are performed on the fused vector. Then, the fused multi-scale features are applied to fault diagnosis.

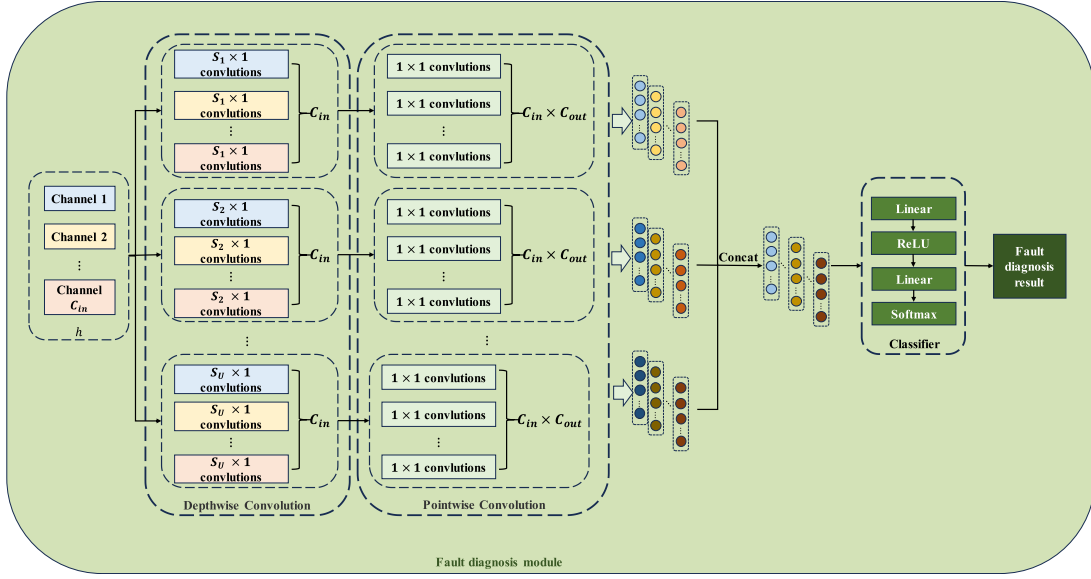


Fig. 3. Flowchart of multi-scale feature extraction and fault diagnosis module.

Algorithm 1: The proposed fault diagnosis algorithm

Input: z_n^t , P_i , the number of iterations $Epoch$, the learning rates of generator α , the fault class labels $y_j, j = 1, 2, \dots, \mathcal{N}$

for $i = 1$ to $Epoch$ **do**

 Get $\hat{P}_i$ by P_i segmentation with sliding windows

$W \leftarrow \text{Eq.}(9)\text{--Eq.}(10)$

$z^i \leftarrow \text{Eq.}(11)$

$h \leftarrow \text{Eq.}(15)\text{--Eq.}(16)$

$\hat{f}_C^{(u)} \leftarrow \text{Eq.}(17)\text{--Eq.}(18)$

$F \leftarrow F = (\hat{f}_C^{(1)}, \hat{f}_C^{(2)}, \dots, \hat{f}_C^{(U)})$

 Fault diagnosis result $\hat{y}_j \leftarrow \text{Eq.}(19)$

$L = -\sum_{j=1}^{\mathcal{N}} y_j \log(\hat{y}_j)$

$\theta_D \leftarrow \theta_D - \alpha \frac{\partial L}{\partial \theta_D}, \theta_F \leftarrow \theta_F - \alpha \frac{\partial L}{\partial \theta_F}, \theta_S \leftarrow \theta_S - \alpha \frac{\partial L}{\partial \theta_S}$

end

Output: Parameters of optimal region selection module θ_S , parameters of the unified representation and fusion module θ_F , and parameters of multi-scale feature extraction and fault diagnosis module θ_D

4. Industrial applications to the HRP

In this section, the dataset is constructed using actual data from HRP, and comparative experiments are set up to verify the effectiveness and superiority of the proposed framework.

4.1. Background of the HRP

As shown in Fig. 4, as a metal working process, hot rolling is widely used in the steel industry. The HRP mainly includes furnaces, rolling mill process, finishing mill process, and laminar cooling. Those processes are interrelated and together form a complex process with strong continuity and high coupling.

Finishing mill process, as a critical stage in the hot rolling process, enables precise dimensional control through continuous rolling with multiple mills. The finishing rolling process is complex and delicate, involving numerous control loops and subsystems with highly coupled relationships. When a fault occurs in the finishing rolling process, it propagates through the coupled systems, leading to anomalies in multiple sensor data. Therefore, to fully characterize fault features

Table 1

The optimal region select module for fault image data.

Module name	Module architectures	Input size	Output size
Optimal region selection module	$W_Q : \text{Linear}(S_w \times S_w, S_w \times S_w)$ $W_K : \text{Linear}(S_w \times S_w, S_w \times S_w)$	$N_p \times S_w^2$	$\hat{P}_i : S_w \times S_w$

Table 2

The unified representation and fusion module.

Module name	Module architectures	Input size	Output size
Unified representation and fusion module	$w_1^t : \text{Tensor}(r, 784, 128)$	$z_n^t : 1 \times N \times 784$ $z_m^t : 1 \times M \times S_w^2$	$h' : 1 \times r \times 128$ $h' : 1 \times r \times 128$
	$w_2^t : \text{Tensor}(r, 784, 128)$		
	$w_3^t : \text{Tensor}(r, 784, 128)$		
	$w_1^t : \text{Tensor}(r, S_w^2, 128)$		
	$w_2^t : \text{Tensor}(r, S_w^2, 128)$		

and achieve accurate fault diagnosis, it is necessary to integrate data in different sources for comprehensive analysis. However, the types of fault data captured by different sensors are usually diverse, such as time series data, image data, etc., with significant differences in dimensions and structures, making it difficult to achieve multi-source heterogeneous data fusion. Therefore, researching unified representation and fusion methods for multi-source heterogeneous fault data in the finishing rolling process is of great significance for improving the accuracy of fault diagnosis.

4.2. Dataset and model construction

Actual data from HRP is used to construct the dataset to verify the effectiveness of the proposed framework. Firstly, the random forest algorithm [33] is applied to select different data sources in the HRP. By calculating the importance of the data from different sources and ranking them, the top three data sources with the highest importance are selected. Subsequently, the dataset is constructed with the data from those data sources, which is split into a training set and a test set. In the training set, for each fault class, there are respectively 1,000 time series data and image data samples, while in the test set, there are 200 time series data and image data samples for each fault class. In addition, each time series data sample has a length of 784, and the image data sample has a dimension of 28×28 .

The structure of three modules constructed is illustrated in Tables 1, 2, and 3. Of which, S_w is the sliding window size which is set to 10, N_p

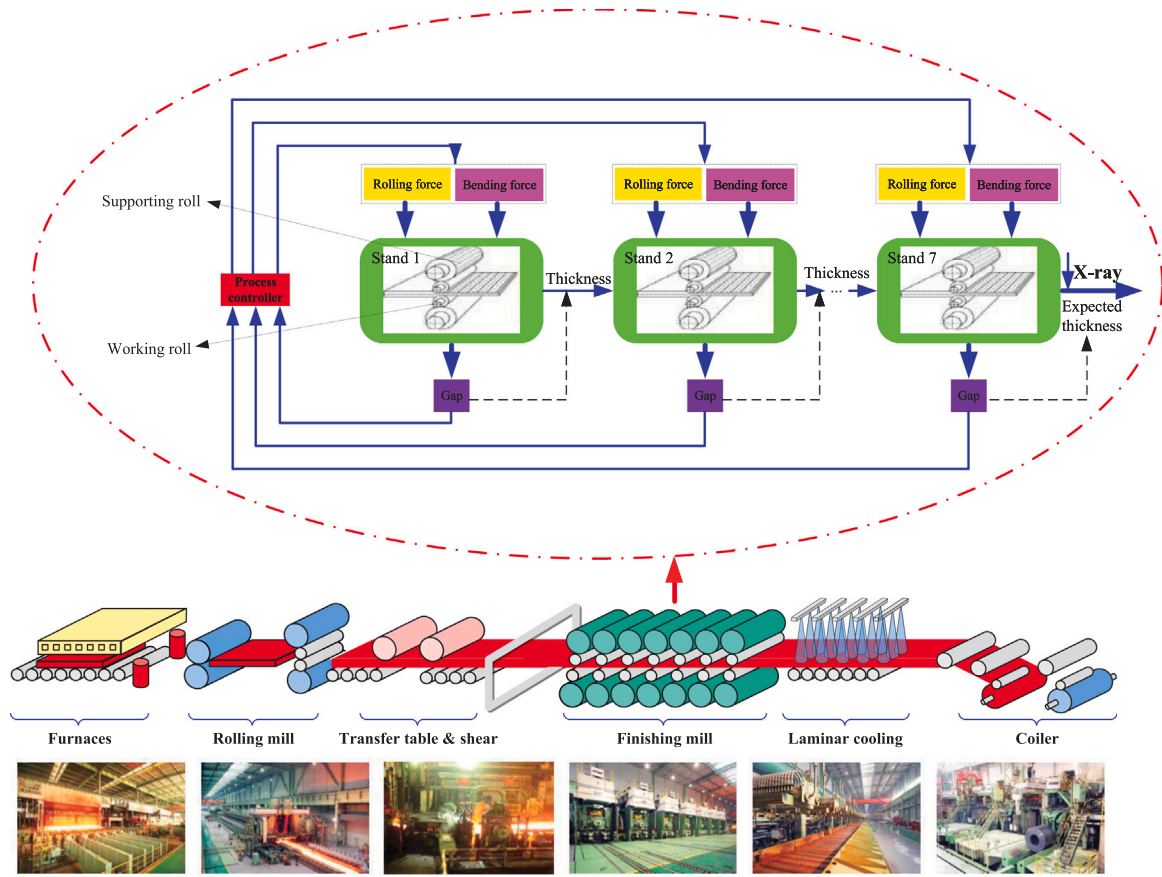


Fig. 4. Schematic layout of the HRP.

Table 3
The multi-scale feature extraction and fault diagnosis module.

Module name	Module architectures	Input size	Output size
Multi-scale feature extraction and fault diagnosis module	Conv1d(r , 16, 3, 1, 1)	$h : 1 \times r \times 128$	$1 \times l$
	ReLU, BN		
	MaxPool1d(2, 2)		
	Dropout		
	Conv1d(r , 16, 5, 1, 2)		
	ReLU, BN		
	MaxPool1d(2, 2)		
	Dropout		
	Conv1d(r , 16, 7, 1, 3)		
	ReLU, BN		
	MaxPool1d(2, 2)		
	Dropout		
	Linear(3072, 128)		
	ReLU		
	Linear(128, l)		
	Softmax		

is the number of image patches after segmentation of fault image data which is 19, and l is the number of classifications which are 3 in case 1 and 4 in case 2. Furthermore, the rank r in the proposed framework is set to 3.

4.3. Case 1

To verify the effectiveness and superiority of the proposed method, fault diagnosis experiments with three classifications are set up. In the

experiments, the epoch and batch size are set to 60 and 20, respectively, and the Adam optimizer is selected for optimization [34]. In addition, those classifications are the normal state, the rolling force fault in the 1st stand, and the rolling gap value fault in the 1st stand. The details of the samples are shown in Table 4.

4.3.1. Ablation experiments

In order to select the appropriate hyperparameters for better model training performance, two sets of ablation experiments are conducted for the learning rate and weight decay factor during model training:

(1) The weight decay factor is kept as 0, and the model is trained while the learning rates are respectively set to 0.001, 0.0001, and 0.00001. The appropriate learning rate is selected by comparing the model performance under different learning rates;

(2) The learning rate that achieves better model performance from above step is selected, and the model is trained with weight decay factors of 0.001, 0.01, and 0.03, to compare the model performance under different weight decay factors.

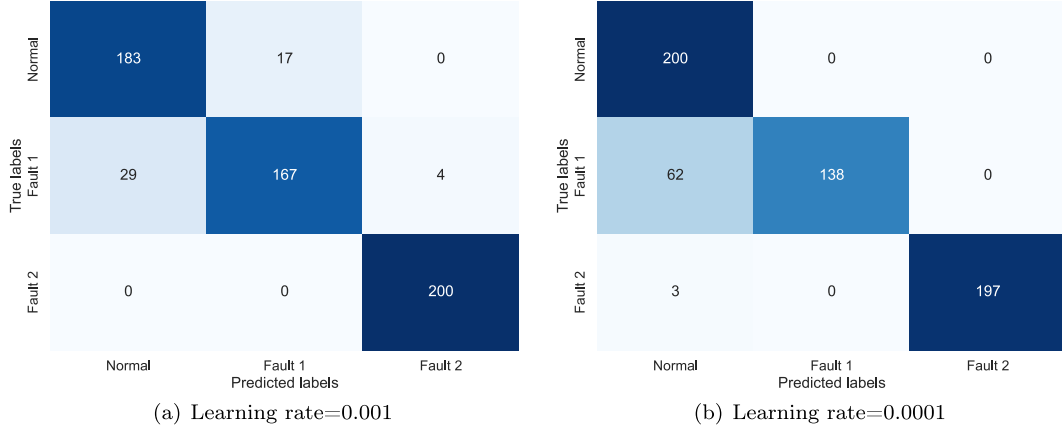
With the learning rate respectively set to 0.001, 0.0001, and 0.00001, the model is trained using the data from training set, while the weight decay factor is kept as 0. Then, the data from the test set is used to test the model.

From the results of Table 5 and Fig. 5, it can be observed that the best fault diagnosis performance of the trained model is achieved at the learning rate of 0.00001, which is higher in accuracy, precision, recall, and F1 score than that when the learning rate is 0.001, 0.0001. Therefore, 0.00001 is selected as the learning rate to train the model. Further, the model is trained at a weight decay factor of 0.001, 0.01,

Table 4

The details of the samples in case 1.

Classifications	Fault severities	Fault locations	Training samples	Testing samples	Labels
Normal	–	–	1000	200	0
Rolling force fault	Serious	The 1st stand	1000	200	1
Rolling gap value fault	Serious	The 1st stand	1000	200	2

**Fig. 5.** Confusion matrixes of testing results with different learning rates in case 1.**Table 5**

Ablation experiment results of different learning rates in case 1.

Learning rates	Accuracy	Precision	Recall	F1 score
0.001	0.9167	0.9171	0.9167	0.9161
0.0001	0.8917	0.9182	0.8917	0.8897
0.00001	0.9317	0.9425	0.9317	0.9319

Table 6

Ablation experiment results of different weight decay factors in case 1.

Weight decay factors	Accuracy	Precision	Recall	F1 score
0.001	0.8650	0.8769	0.8650	0.8643
0.01	0.9217	0.9353	0.9217	0.9219
0.03	0.9733	0.9753	0.9733	0.9733

and 0.03, respectively. The trained model is then tested with the data from test set.

On the basis of the experimental results in Table 6 and Fig. 6, when the learning rate is 0.00001, the best fault diagnosis performance is achieved. Furthermore, the model is trained with weight decay factors of 0.001, 0.01, and 0.03 to alleviate the overfitting issue during the training processes. When the weight decay factor is 0.03, the fault diagnosis performance is relatively optimal, with all four metrics higher

than the model performance with weight decay factors of 0.001 and 0.01. Therefore, it is more reasonable to set the learning rate and weight decay factor to 0.00001 and 0.03 respectively during the model training processes.

4.3.2. Simulation results and discussions

In this section, the proposed method is compared in terms of fault diagnosis performance and model robustness with the following methods, and the model complexity is analysed. In addition, based on the results of ablation experiments, both the learning rate and weight decay factor during the training processes are set to 0.00001 and 0.03, respectively.

(1) Unified representation and fusion method of multi-source heterogeneous data (URF). This method directly fuses fault image data with time series data without performing optimal region selection on the fault images, and the fusion vector is then used for fault diagnosis.

(2) Feature level fusion based method of multi-source heterogeneous data (FLF). This method independently preforms feature extraction on time series and image data using one-dimensional CNN (1DCNN) and two-dimensional CNN (2DCNN) [16,17], respectively, and those features are then fused through concatenation for fault diagnosis.

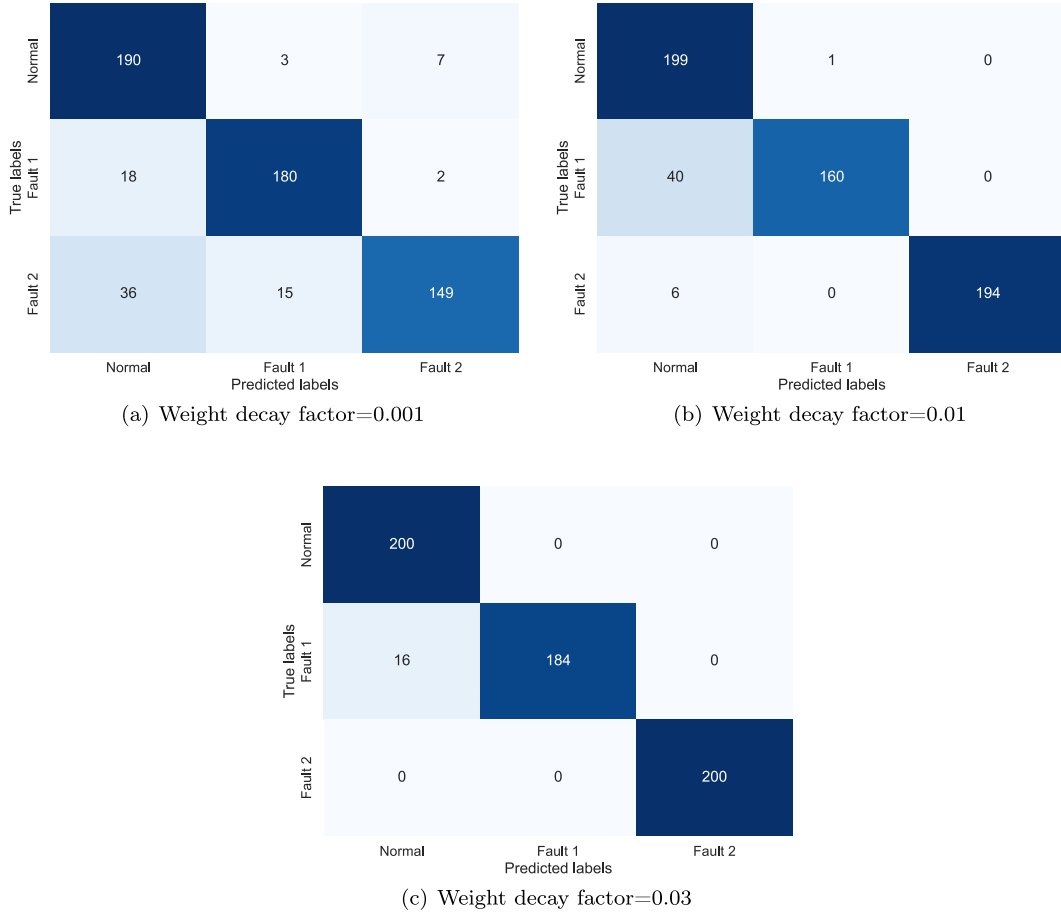


Fig. 6. Confusion matrixes of testing results with different weight decay factors in case 1.

(3) 1DCNN based feature extraction and fault diagnosis method (1DCNN-FEFD). This method uses the traditional 1DCNN for feature extraction and fault diagnosis after obtaining the unified representation of multi-source heterogeneous data [17].

(4) The attention-enhanced information fusion diagnosis network (AIFN) [29]. This method converts the time series data into image data and performs data level fusion with image data, which is then used for feature extraction and fault diagnosis.

(5) The deep feature interactive network (DFINet) [28]. This method extracts the public and private features from multi-source heterogeneous data through the feature interaction module, which are then fused for fault diagnosis.

The trained fault diagnosis models are firstly tested using the testing set data, and the fault diagnosis performance is evaluated by accuracy, precision, recall, and F1 score, to compare the fault diagnosis performance of different methods. The fault diagnosis results are shown in Table 7. In addition, the feature scatter plots obtained by visualizing the features input to the classifier through the t-distributed stochastic neighbour embedding (t-SNE) dimensionality reduction and the confusion matrix of the diagnosis results are presented in Figs. 7 and 8, respectively.

Noise is a key factor affecting the fault diagnosis performance. In order to test the robustness of different methods, noise is added to the testing set data, which are subsequently used to test the trained fault diagnosis model to compare the fault diagnosis performance of different methods with different signal-to-noise ratio (SNR) [11]. The definition of SNR is shown as:

$$SNR = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right) \quad (20)$$

Table 7

Fault diagnosis results of different methods in case 1.

Methods	Accuracy	Precision	Recall	F1 score
URF	0.9183	0.9265	0.9183	0.9194
FLF	0.9233	0.9364	0.9233	0.9232
1DCNN-FEFD	0.9383	0.9480	0.9383	0.9384
AIFN	0.9167	0.9189	0.9167	0.9168
DFINet	0.9733	0.9753	0.9733	0.9735
The proposed method	0.9733	0.9753	0.9733	0.9733

where p_{signal} is the power of signal, and p_{noise} is the power of noise. Robustness tests on different methods are conducted under three scenarios: adding noise only to the time series data, adding noise only to the image data, and adding noise to both the time series and image data simultaneously. The test results are shown in Table 8, Table 9, and Table 10, respectively.

Model lightweighting is significant for reducing computational cost and improving the real-time performance of fault diagnosis. In order to verify the model complexity of the fault image data optimal region selection module and the multi-scale convolution based lightweight CNN module, the former is compared with the fault diagnosis model without the application of the optimal region selection module, and the latter is compared with the multi-scale convolution based traditional CNN module. Additionally, the lightweighting effect of both modules is validated through FLOPs and the number of parameters [35]. The experiment results are displayed in Table 11.

Based on the above experimental results, it can be seen that the proposed method outperforms other ones in terms of fault diagnosis performance. The URF method, compared to the proposed method, shows reductions of 0.055, 0.0488, 0.055, and 0.0539 in the four

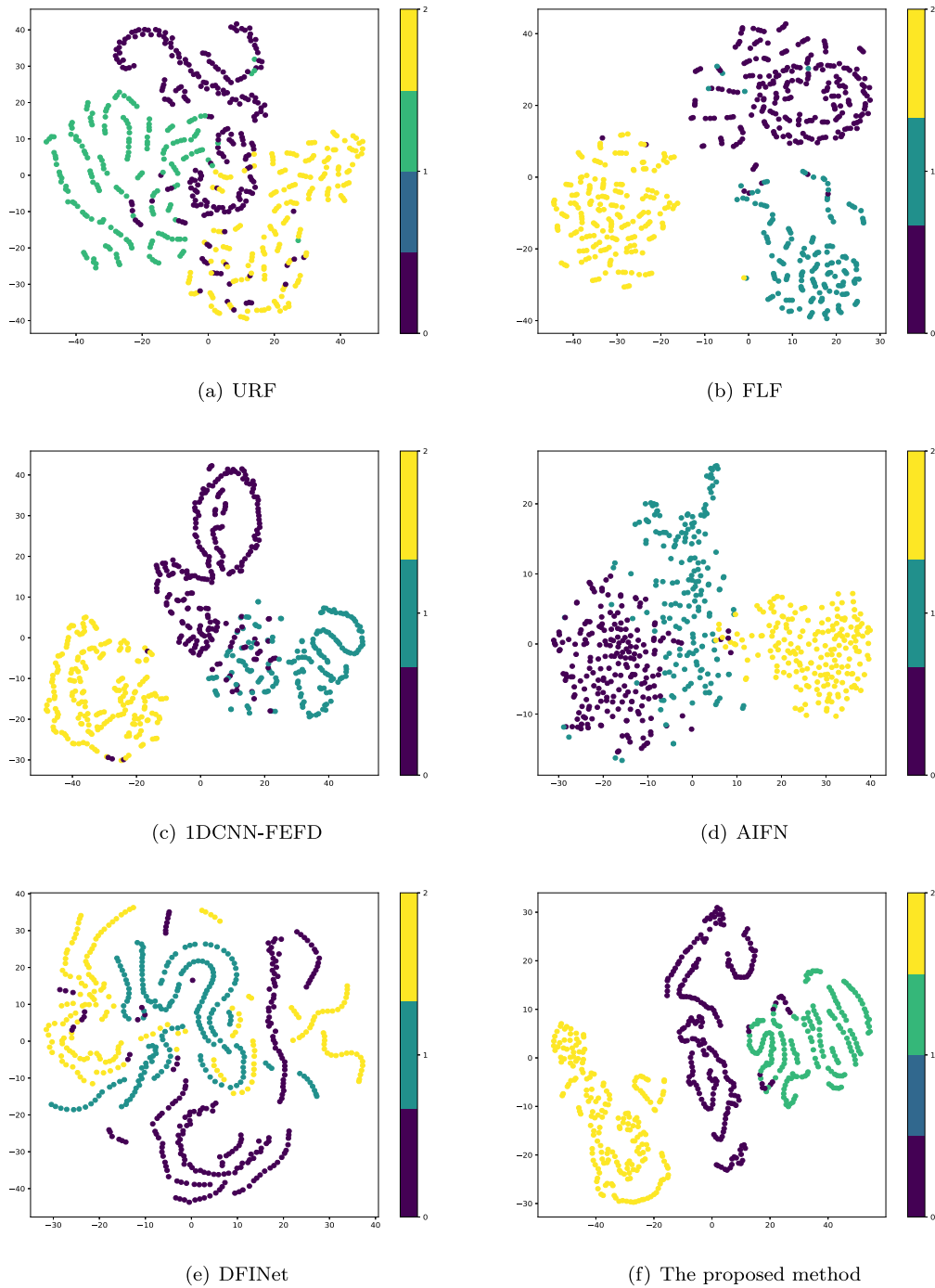


Fig. 7. Scatter plots for feature distribution of different methods in case 1.

metrics, indicating that performing optimal region selection on image data effectively reduces the interference of redundant information in the images, thereby improving fault diagnosis performance. In the FLF method, due to the potential information loss during the feature extraction process, the fault diagnosis performance is lower than that of the proposed method, with reductions of 0.05, 0.0389, 0.05, and 0.0501 in the four metrics. Additionally, when faced with noise interference, the fault diagnosis performance decreases significantly, indicating poor robustness of the method. Compared to the proposed method, the fault diagnosis results of the 1DCNN-FEFD method show reductions of 0.035,

0.0273, 0.035, and 0.0349 in the four metrics. It can be indicated that, for the proposed method, multi-scale features from the fused vector representation can be effectively extracted, improving the fault diagnosis performance.

The AIFN method, compared to the proposed method, shows a reduction of 0.0557, 0.0564, 0.0557, and 0.0565 in the four fault diagnosis metrics, respectively. This suggests that there may be the problem of information loss when converting time series data into image data, making it difficult for the extracted features to fully reflect fault information, thereby affecting the fault diagnosis performance.

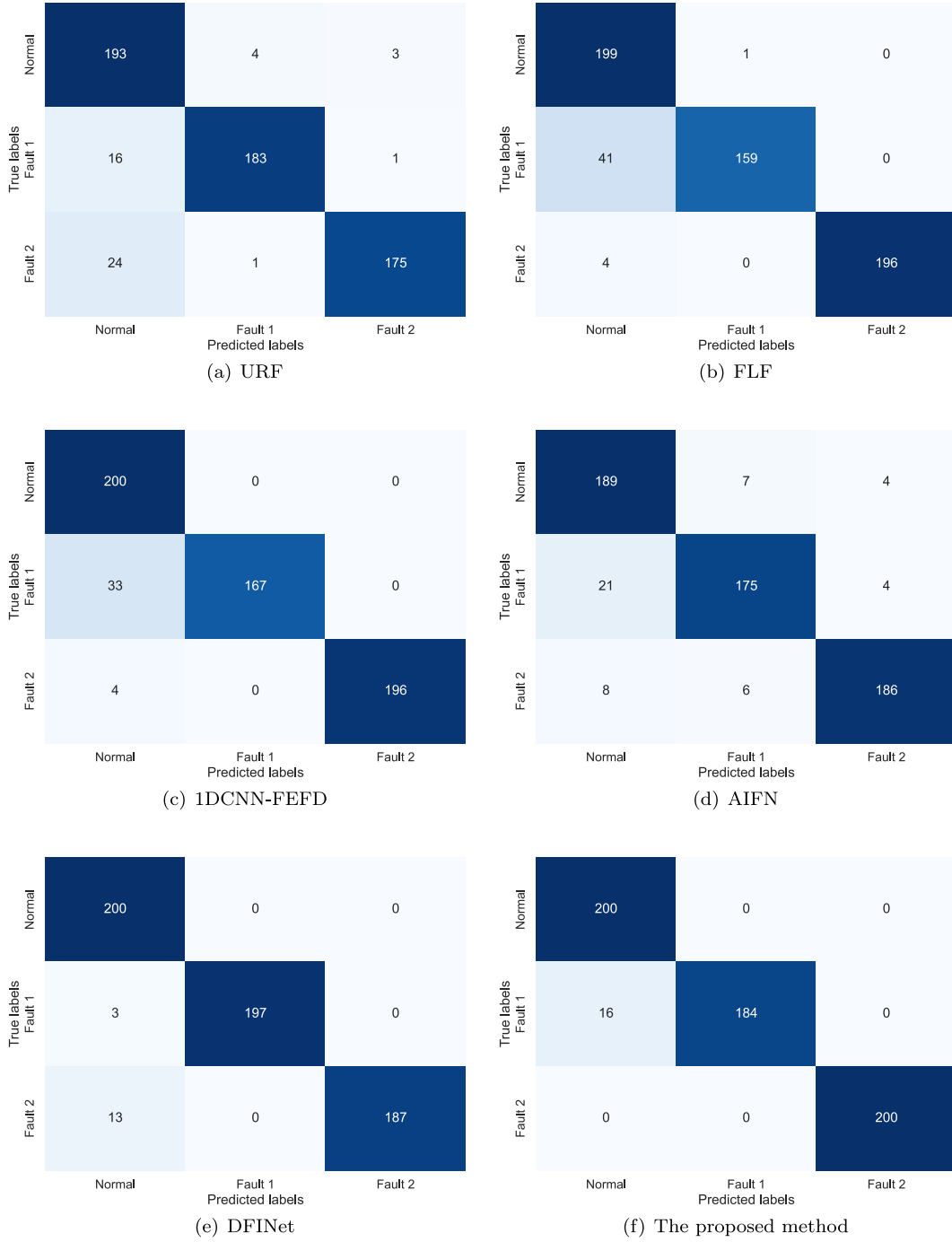


Fig. 8. Confusion matrixes for testing results of different methods in case 1.

DFNet can reduce the information redundancy through the extraction of private and public features to improve the fault diagnosis accuracy, and its fault diagnosis performance is the same as the proposed method with a slight advantage in the F1 score. However, when the input data contains noise, the fault diagnosis performance decreases significantly, with a maximum decrease of 0.8068. It indicates that the method is less robust and struggles to achieve accurate fault diagnosis in the presence of noise interference. In addition, in terms of model complexity, the optimal region selection module increases the FLOPs, but effectively reduces the number of parameters. The multi-scale convolution based lightweight CNN module reduces both the FLOPs and the number of model parameters through depthwise separable convolution, realizing the lightweight. Therefore, based on the above experimental results and

analysis, the proposed method demonstrates superior fault diagnosis performance in HRP.

4.4. Case 2

In this section, four classification experiments with the same hyper-parameters and optimizer settings as in case 1 are set up, where the classifications are the normal state, the rolling force fault in the 1st stand, the rolling gap value fault in the 1st stand, and the bend rolling force fault in the 5th stand, to further validate the effectiveness and generalization of the proposed method. The details of samples are given in Table 12.

Table 8

Robustness test results for time series data with noises in case 1.

SNR	Methods	URF	FLF	1DCNN-FEFD	AIFN	DFINet	The proposed method
2 dB	Accuracy	0.8933	0.7917	0.9133	0.7700	0.3750	0.9650
	Precision	0.9051	0.8718	0.9306	0.8294	0.4493	0.9683
	Recall	0.8933	0.7917	0.9133	0.7700	0.3750	0.9650
	F1 score	0.8951	0.7948	0.9131	0.7690	0.2461	0.9650
1 dB	Accuracy	0.8833	0.7750	0.9100	0.7383	0.3617	0.9633
	Precision	0.8966	0.8657	0.9285	0.8087	0.4477	0.9670
	Recall	0.8833	0.7750	0.9100	0.7383	0.3617	0.9633
	F1 score	0.8853	0.7786	0.9096	0.7360	0.2225	0.9633
−1 dB	Accuracy	0.8717	0.6817	0.9033	0.6283	0.3433	0.9633
	Precision	0.8877	0.8372	0.9229	0.7550	0.4456	0.9670
	Recall	0.8717	0.6817	0.9033	0.6283	0.3433	0.9633
	F1 score	0.8740	0.6825	0.9030	0.6167	0.1873	0.9633
−2 dB	Accuracy	0.8583	0.6050	0.9017	0.5683	0.3350	0.9617
	Precision	0.8745	0.8159	0.9218	0.7124	0.4446	0.9650
	Recall	0.8583	0.6050	0.9017	0.5683	0.3350	0.9617
	F1 score	0.8608	0.5943	0.9015	0.5440	0.1702	0.9617

Table 9

Robustness test results for image data with noises in case 1.

SNR	Methods	URF	FLF	1DCNN-FEFD	AIFN	DFINet	The proposed method
2 dB	Accuracy	0.8917	0.8500	0.9200	0.6233	0.4300	0.9717
	Precision	0.8998	0.8966	0.9355	0.7879	0.3657	0.9739
	Recall	0.8917	0.8500	0.9200	0.6233	0.4300	0.9717
	F1 score	0.8931	0.8505	0.9197	0.5214	0.3325	0.9716
1 dB	Accuracy	0.8867	0.8317	0.9083	0.6117	0.3967	0.9733
	Precision	0.8949	0.8882	0.9281	0.4542	0.3415	0.9753
	Recall	0.8867	0.8317	0.9083	0.6117	0.3967	0.9733
	F1 score	0.8880	0.8321	0.9079	0.5039	0.2877	0.9733
−1 dB	Accuracy	0.8767	0.7983	0.9033	0.5933	0.3633	0.9750
	Precision	0.8853	0.8744	0.9221	0.4581	0.3011	0.9767
	Recall	0.8767	0.7983	0.9033	0.5933	0.3633	0.9750
	F1 score	0.8782	0.8003	0.9031	0.4919	0.2384	0.9750
−2 dB	Accuracy	0.8667	0.7717	0.9117	0.5833	0.3467	0.9783
	Precision	0.8780	0.8645	0.9275	0.4563	0.2695	0.9797
	Recall	0.8667	0.7717	0.9117	0.5833	0.3467	0.9783
	F1 score	0.8685	0.7735	0.9117	0.4839	0.2106	0.9783

Table 10

Robustness test results for both time series and image data with noises in case 1.

SNR	Methods	URF	FLF	1DCNN-FEFD	AIFN	DFINet	The proposed method
2 dB	Accuracy	0.8667	0.5933	0.9067	0.6117	0.3700	0.9700
	Precision	0.8829	0.8123	0.9271	0.7844	0.4487	0.9720
	Recall	0.8667	0.5933	0.9067	0.6117	0.3700	0.9700
	F1 score	0.8694	0.5507	0.9059	0.5318	0.2374	0.9700
1 dB	Accuracy	0.8535	0.5117	0.9050	0.5817	0.3550	0.9717
	Precision	0.8738	0.8019	0.9254	0.7762	0.4469	0.9739
	Recall	0.8535	0.5117	0.9050	0.5817	0.3550	0.9717
	F1 score	0.8609	0.4338	0.9044	0.5012	0.2101	0.9716
−1 dB	Accuracy	0.8200	0.3683	0.9017	0.4950	0.3400	0.9683
	Precision	0.8343	0.4485	0.9156	0.7491	0.4452	0.9705
	Recall	0.8200	0.3683	0.9017	0.4950	0.3400	0.9683
	F1 score	0.8229	0.2345	0.9017	0.4056	0.1806	0.9683
−2 dB	Accuracy	0.7883	0.3417	0.8950	0.4683	0.3333	0.9700
	Precision	0.8008	0.4454	0.9064	0.7436	0.1111	0.9720
	Recall	0.7883	0.3417	0.8950	0.4683	0.3333	0.9700
	F1 score	0.7912	0.1840	0.8945	0.3777	0.1667	0.9700

Table 11

Comparison results of model complexity in case 1.

Module	FLOPs	Parameters
Without the optimal region selection module	5640 236	1918 145
With the optimal region selection module	32 598 332	102 125
Traditional CNN module	84 528	648
Lightweight CNN module	36 804	270

4.4.1. Ablation experiments

To select the appropriate learning rate as well as the weight decay factor and improve the performance of model training, two sets of ablation experiments for the learning rate and the weight decay factor are set up, respectively. Both sets of ablation experiments are set up in the same way as case 1.

Firstly, the learning rate is respectively set to 0.001, 0.0001, and 0.00001, while the weight decay factor is set to 0, to train and test the model. The test results are shown in Table 13 and Fig. 9.

Table 12
The details of the samples in case 2.

Classifications	Fault severities	Fault locations	Training samples	Testing samples	Labels
Normal	–	–	1000	200	0
Rolling force fault	Serious	The 1st stand	1000	200	1
Rolling gap value fault	Serious	The 1st stand	1000	200	2
Bend rolling force fault	Serious	The 5th stand	1000	200	3

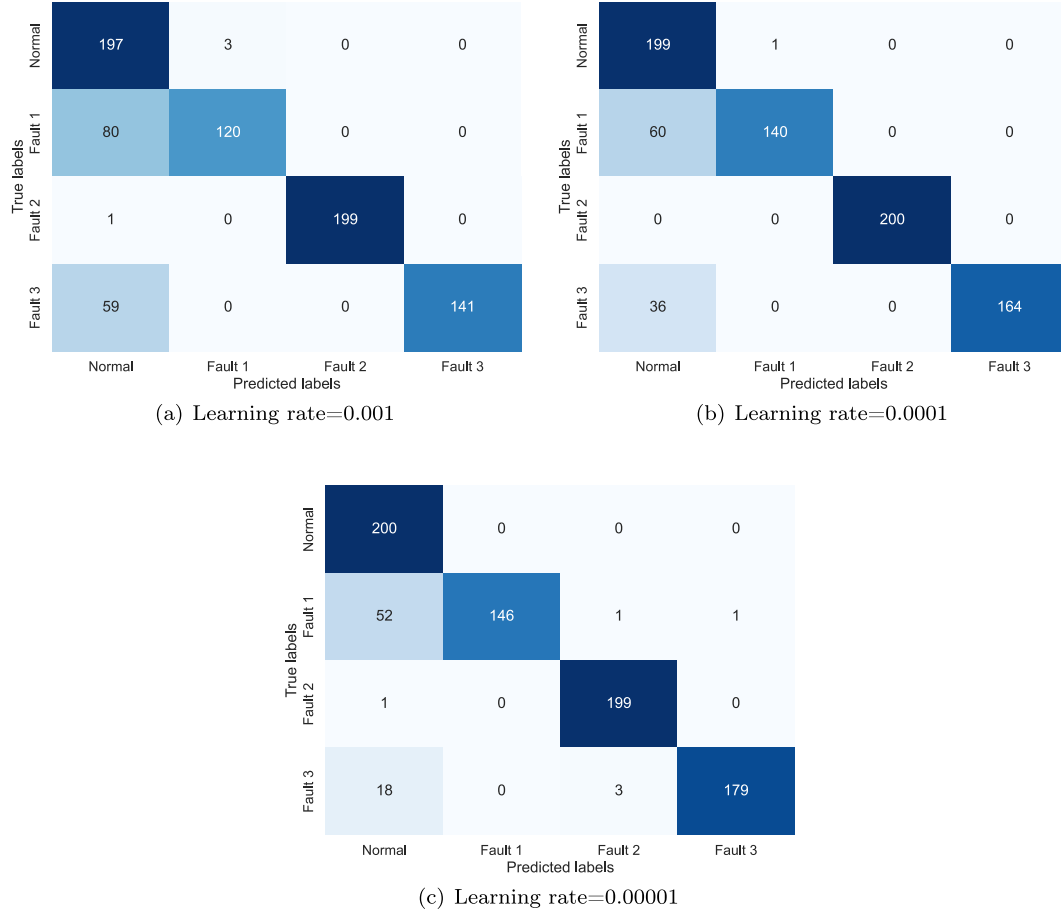


Fig. 9. Confusion matrixes of testing results with different learning rates in case 2.

Table 13
Ablation experiment results of different learning rates in case 2.

Learning rates	Accuracy	Precision	Recall	F1 score
0.001	0.8213	0.8900	0.8213	0.8253
0.0001	0.8788	0.9169	0.8787	0.8816
0.00001	0.9050	0.9282	0.9050	0.9057

Table 14
Ablation experiment results of different weight decay factors in case 2.

Weight decay factors	Accuracy	Precision	Recall	F1 score
0.001	0.8725	0.9135	0.8725	0.8720
0.01	0.8912	0.9221	0.8912	0.8930
0.03	0.9712	0.9731	0.9712	0.9708

From the ablation experiments for learning rate, it is clear that when the learning rate is 0.00001, the performance of model training is relatively improved. Therefore, it is more reasonable to set the learning rate to 0.00001. Subsequently, the weight decay factor is set to 0.001, 0.01, 0.03 with the learning rate of 0.00001 to train and test the model. The test results are shown in Table 14 and Fig. 10.

The ablation experiment results of weight decay factor show that the model achieves the highest fault diagnosis accuracy when the weight decay factor is set to 0.3 with the learning rate of 0.00001. It can be respectively improved by 0.0987 and 0.08, compared to the cases of weight decay factors are 0.001 and 0.01. In summary, the performance of the model is more effective when the learning rate is 0.00001 and the weight decay factor is 0.03.

4.4.2. Simulation results and discussions

Similar to case 1, the proposed method is compared with five methods. And the learning rate and weight decay factor for model training are set to 0.00001 and 0.03, respectively.

Firstly, in order to test the fault diagnosis performance of different methods, the models are tested using the testing set data and evaluated by the same four metrics as in case 1. The test results are shown in Table 15, Fig. 11, and Fig. 12.

The fault diagnosis performance of different methods is tested by adding noise to the testing data to verify the robustness of models, and the results are listed in Tables 16, 17, and 18.

In order to verify the change of model complexity when the fault class increases, the FLOPs and model parametric quantities of the optimal region selection module and the multi-scale based lightweight

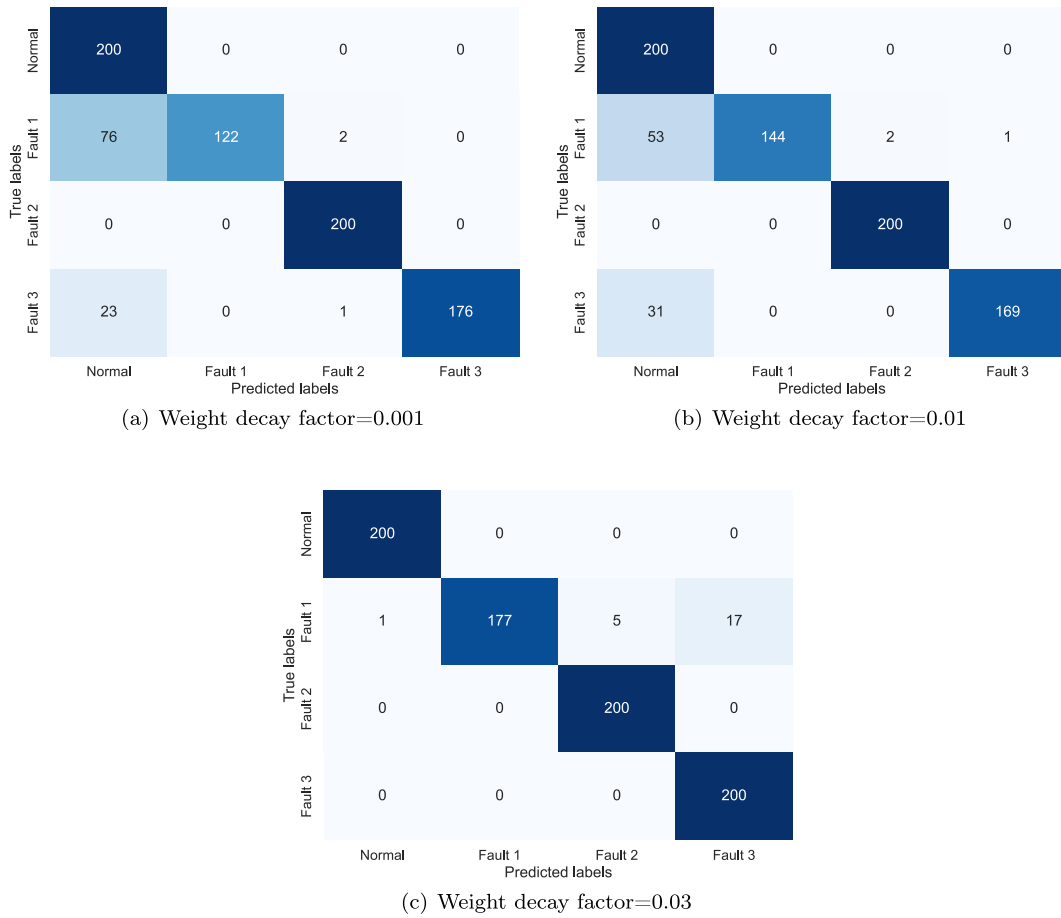


Fig. 10. Confusion matrixes of testing results with different weight decay factors in case 2.

Table 15

Fault diagnosis results of different methods in case 2.

Methods	Accuracy	Precision	Recall	F1 score
URF	0.9387	0.9431	0.9387	0.9386
FLF	0.9525	0.9594	0.9525	0.9520
1DCNN-FEFD	0.9175	0.9333	0.9175	0.9150
AIFN	0.9087	0.9108	0.9088	0.9087
DFINet	0.9275	0.9352	0.9275	0.9256
The proposed method	0.9712	0.9731	0.9712	0.9708

CNN module are tested in the same way as case 1, and the results are presented in Table 19.

Based on the above experimental results, the proposed method is more superior compared to the other methods. When the image data is analysed directly and fused with the time series data for fault diagnosis without optimal region selection, both the fault diagnosis performance and robustness are reduced. It can be indicated that performing optimal region selection on image data can effectively reduce redundant information and avoid interference of background noise in the images. Compared to the FLF method, the proposed method shows improvements in all four fault diagnosis metrics and demonstrates stronger robustness, indicating that it can mitigate the problem of information loss during fusion and enhance data utilization. When the traditional 1DCNN is applied to feature extraction and fault diagnosis based on the fused vector representation of time series and image data, the comprehensive fault diagnosis performance is reduced due to the difficulty of obtaining multi-scale features by single-scale convolutions. In the AIFN method, information loss may occur during the process of converting time series data into image data, which leads to insufficient information after data fusion. As a result, the fault diagnosis performance and

robustness are lower than that of the proposed method. Compared to the proposed method, fault diagnosis performance of DFNet decreases by 0.0437, 0.0379, 0.0437, and 0.0452, respectively. Additionally, the fault diagnosis performance decreases with a range from 0.1384 to 0.7286 when affected by noise interference. It indicates that DFNet suffers from insufficient feature extraction during the feature extraction process, which affects the fault diagnosis performance. In addition, in terms of model complexity, although the overall FLOPs as well as the parameters increase as the fault classes increase, the model still demonstrates excellent lightweight performance. It can be concluded that the comprehensive fault diagnosis performance of the proposed method is better than that of other methods.

From case 1 and case 2, it is clear that the proposed method can effectively select the optimal region containing useful information from the fault image data in the HRP, thereby reducing interference from redundant information in the image data. Moreover, through the unified representation and fusion module, the challenge of fusing multi-source heterogeneous data, which has significant differences in dimensions and structures, is addressed, thus improving the data information utilization and preventing information loss. Additionally, through the multi-scale feature extraction and fault diagnosis module, multi-scale features can be captured in the fused vector representation, further enhancing the fault feature representation capability, which provides a feasible solution for fault diagnosis based on multi-source heterogeneous data fusion in industrial processes.

5. Conclusion

In this paper, in response to the challenges of integrating multi-source heterogeneous data with significant differences in dimensions

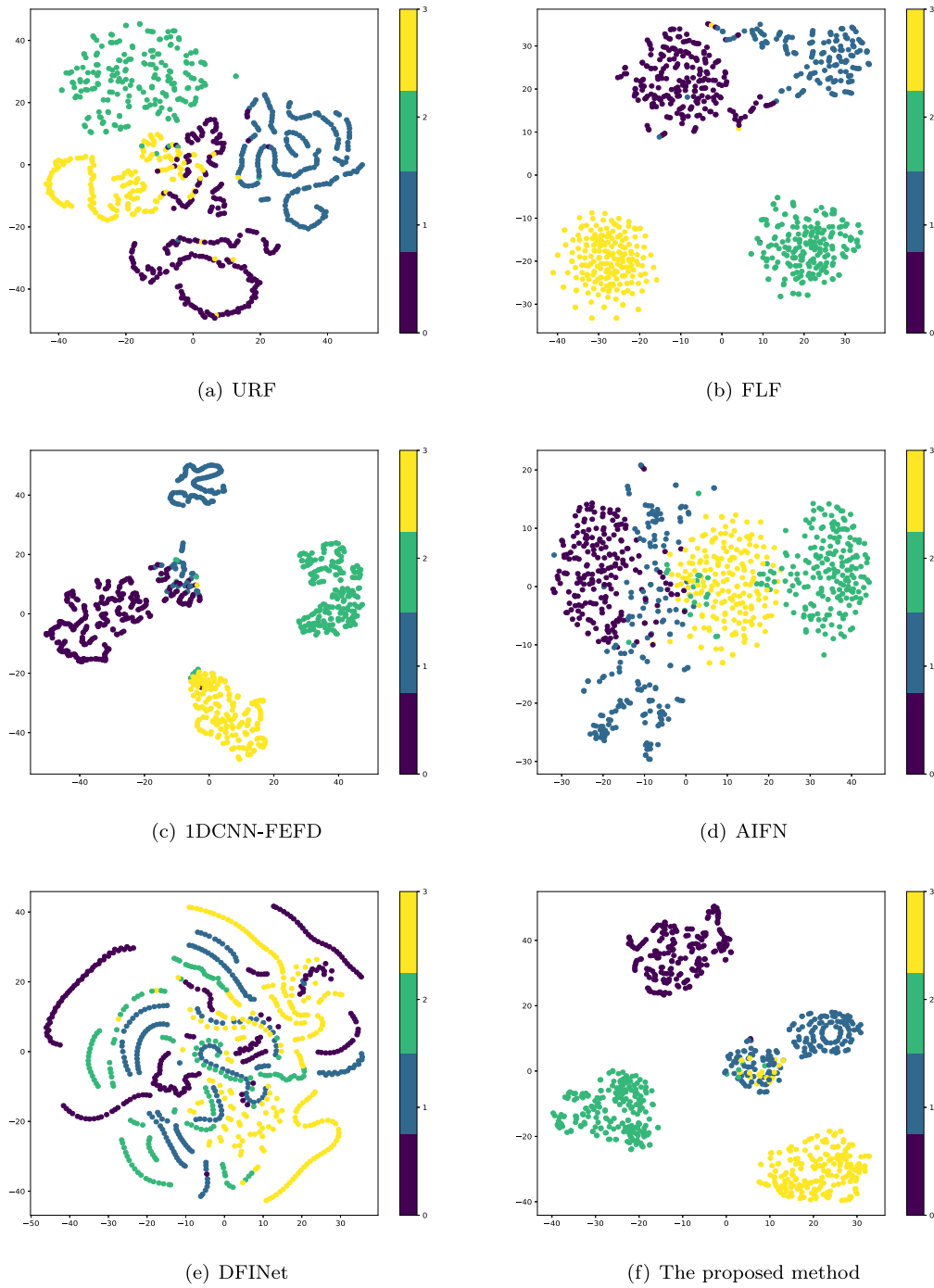


Fig. 11. Scatter plots for feature distribution of different methods in case 2.

and structure in industrial processes, and to avoid potential information loss during the data fusion processes, a unified representation and fusion framework of multi-source heterogeneous data is proposed for fault diagnosis. The fault image data is firstly divided into image patches through a sliding window, whose weights are then calculated through the self-attention mechanism based module. Subsequently, the image patch with the highest importance weight is selected as the optimal representative region of image data. Afterwards, tensor fusion is performed on the time series data and the selected image data region,

and low-rank decomposition is utilized to obtain a low-dimensional unified vector representation, thereby achieving the multi-source heterogeneous data fusion. Finally, the fused vector representation is input into a multi-scale lightweight CNN module for multi-scale feature extraction and fault diagnosis. Moreover, two sets of comparative experiments are conducted to verify the effectiveness and superiority of the proposed method in terms of fault diagnosis performance, model robustness, and complexity. Four metrics: accuracy, precision, recall, and F1 score are used to evaluate the performance of different fault

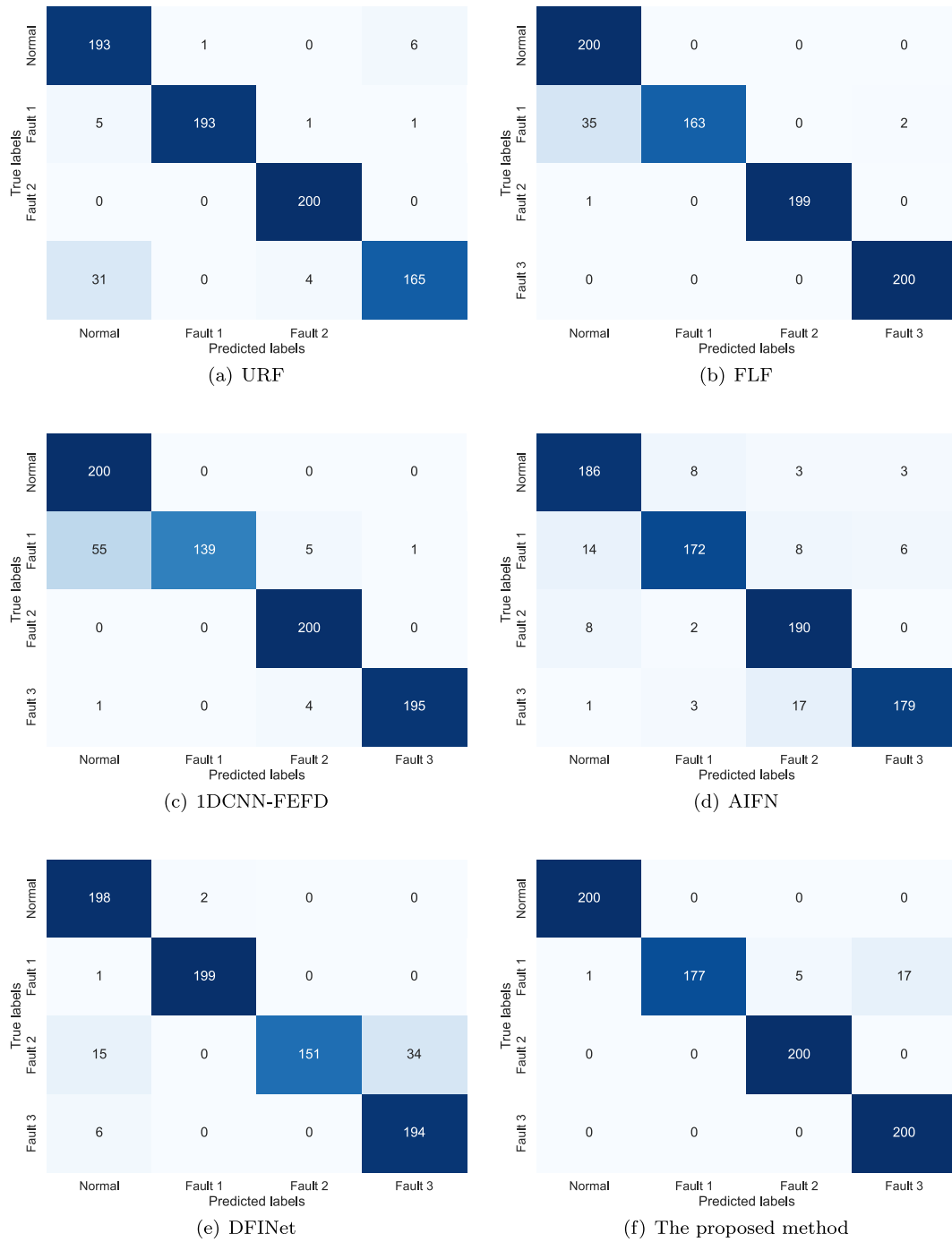


Fig. 12. Confusion matrixes for testing results of different methods in case 2.

Table 16
Robustness test results for time series data with noises in case 2.

SNR	Methods	URF	FLF	1DCNN-FEFD	AIFN	DFINet	The proposed method
2 dB	Accuracy	0.9250	0.8450	0.9113	0.7875	0.3875	0.9650
	Precision	0.9322	0.9021	0.9282	0.8190	0.2992	0.9670
	Recall	0.9250	0.8450	0.9113	0.7875	0.3875	0.9650
	F1 score	0.9254	0.8438	0.9086	0.7659	0.2847	0.9646
1 dB	Accuracy	0.9213	0.8237	0.9100	0.7775	0.3725	0.9575
	Precision	0.9296	0.8942	0.9271	0.8123	0.3053	0.9604
	Recall	0.9213	0.8237	0.9100	0.7775	0.3725	0.9575
	F1 score	0.9216	0.8215	0.9073	0.7560	0.2724	0.9563
−1 dB	Accuracy	0.9000	0.7450	0.9000	0.7525	0.3225	0.9525
	Precision	0.9092	0.8716	0.9204	0.7927	0.2945	0.9554
	Recall	0.9000	0.7450	0.9000	0.7525	0.3225	0.9525
	F1 score	0.9001	0.7481	0.8969	0.7228	0.2167	0.9515
−2 dB	Accuracy	0.8900	0.6700	0.8912	0.7275	0.3075	0.9550
	Precision	0.9010	0.8564	0.9140	0.7805	0.2922	0.9579
	Recall	0.8900	0.6700	0.8912	0.7275	0.3075	0.9550
	F1 score	0.8898	0.6747	0.8881	0.6907	0.1970	0.9541

Table 17
Robustness test results for image data with noises in case 2.

SNR	Methods	URF	FLF	1DCNN-FEFD	AIFN	DFINet	The proposed method
2 dB	Accuracy	0.9313	0.9087	0.9087	0.4025	0.6713	0.9750
	Precision	0.9354	0.9332	0.9257	0.5107	0.7968	0.9764
	Recall	0.9312	0.9087	0.9088	0.4025	0.6713	0.9750
	F1 score	0.9311	0.9073	0.9061	0.2946	0.6394	0.9747
1 dB	Accuracy	0.9250	0.8988	0.9100	0.3638	0.6300	0.9650
	Precision	0.9296	0.9279	0.9267	0.2793	0.7815	0.9670
	Recall	0.9250	0.8987	0.9100	0.3638	0.6300	0.9650
	F1 score	0.9248	0.8970	0.9075	0.2591	0.5842	0.9642
−1 dB	Accuracy	0.9237	0.8850	0.8988	0.2725	0.5075	0.9688
	Precision	0.9283	0.9212	0.9184	0.2785	0.7442	0.9703
	Recall	0.9237	0.8850	0.8988	0.2725	0.5075	0.9688
	F1 score	0.9236	0.8836	0.8958	0.1429	0.3947	0.9681
−2 dB	Accuracy	0.9163	0.8675	0.8912	0.2550	0.4750	0.9762
	Precision	0.9200	0.9134	0.9119	0.3128	0.7349	0.9779
	Recall	0.9163	0.8675	0.8912	0.2550	0.4750	0.9763
	F1 score	0.9162	0.8667	0.8877	0.1102	0.3337	0.9761

Table 18
Robustness test results for both time series and image data with noises in case 2.

SNR	Methods	URF	FLF	1DCNN-FEFD	AIFN	DFINet	The proposed method
2 dB	Accuracy	0.9087	0.6950	0.9000	0.3675	0.3787	0.9613
	Precision	0.9144	0.8626	0.9196	0.5351	0.2988	0.9648
	Recall	0.9087	0.6950	0.9000	0.3675	0.3788	0.9613
	F1 score	0.9087	0.6683	0.8969	0.2680	0.2769	0.9608
1 dB	Accuracy	0.9025	0.5938	0.8962	0.3175	0.3700	0.9663
	Precision	0.9079	0.8452	0.9185	0.5498	0.2985	0.9685
	Recall	0.9025	0.5938	0.8962	0.3175	0.3700	0.9663
	F1 score	0.9025	0.5509	0.8931	0.2118	0.2687	0.9658
−1 dB	Accuracy	0.8838	0.3752	0.8825	0.2562	0.3237	0.9550
	Precision	0.8911	0.5712	0.9052	0.2713	0.2950	0.9583
	Recall	0.8838	0.3752	0.8825	0.2562	0.3237	0.9550
	F1 score	0.8836	0.3072	0.8787	0.1127	0.2183	0.9539
−2 dB	Accuracy	0.8700	0.3088	0.8825	0.2512	0.3075	0.9487
	Precision	0.8779	0.5664	0.9033	0.3126	0.3061	0.9532
	Recall	0.8700	0.3088	0.8825	0.2512	0.3075	0.9487
	F1 score	0.8694	0.2091	0.8785	0.1026	0.1978	0.9478

Table 19
Comparison results of model complexity in case 2.

Module	FLOPs	Parameters
Without the optimal region selection module	5640 271	1918 178
With the optimal region selection module	32 598 367	102 158
Traditional CNN module	84 528	648
Lightweight CNN module	36 804	270

diagnosis models. The experimental results indicate that the fault diagnosis performance of the proposed method is superior to other methods in all four metrics, where high-accuracy fault diagnosis is achieved.

Although the proposed method shows satisfactory performance, it still has some limitations. In the process of selecting the optimal image regions from fault image data, the proposed method firstly divides the image data into patches by a sliding window, where its size and step length are fixed. It may lead to the repetitive selection of non-fault regions, causing computational redundancy. Therefore, re-searching dynamic adjustment methods for the sliding window, which

can adaptively adjust the window and step sizes, is of significant importance for improving the accuracy of fault region selection in fault diagnosis tasks.

CRedit authorship contribution statement

Liang Ma: Writing – review & editing, Supervision, Methodology, Data curation. **Qikai Yang:** Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Kaixiang Peng:** Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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